27. Monday, June 2. Partitions and equivalence relations (7.3)

Definition: Let $\sim$ be an equivalence relation on a set A. Let $a \in A$.

The **equivalence class of a** is the subset

$$[a] = \{ x \in A \mid x \sim a \}$$

Example: Congruence mod 2 on $\mathbb{Z}$: $a \sim b \iff a \equiv b \pmod 2$

$$[0] = \{ a \in \mathbb{Z} \mid a \equiv 0 \pmod 2 \} = \{ \text{even integers} \}$$

$$[1] = \{ a \in \mathbb{Z} \mid a \equiv 1 \pmod 2 \} = \{ \text{odd integers} \}$$

Two classes

Example: $R_=$ The relation on $\mathbb{R}$ where $x R_= y \iff x = y$

Equivalence classes?

For each $a \in \mathbb{R}$, we have

$$[a] = \{ x \in \mathbb{R} \mid x R_= a \} = \{a\}$$

one class for each $a \in \mathbb{R}$.

Goal for Today: If $\sim$ is an equivalence relation on A, Then the equivalence classes partition A.

Theorem 1: Let $\sim$ be an equivalence relation on a nonempty set A. Then

1) $\forall a \in A, \quad a \in [a]$
2) $\forall a, b \in A, \quad a \sim b \iff [a] = [b]$
3) $\forall a, b \in A,$ either $[a] = [b]$ or $[a] \cap [b] = \emptyset$.

Proof: 1) Let $a \in A$. Since $\sim$ is reflexive, $a \sim a$. Therefore $a \in [a] = \{x \in A \mid x \sim a\}$

2) [$\Rightarrow$] Suppose $a, b \in A$ with $a \sim b$. We wish to show $[a] = [b]$.

[$\subseteq$] Let $x \in [a]$. Then $x \sim a$. Since $a \sim b$, by transitivity, we have $x \sim b$. Hence $x \in [b]$.

[$\supseteq$] Exercise. Reverse the roles of a and b, use reflexivity.

[$\Leftarrow$] Suppose $[a] = [b]$. By part (1), we have $a \in [a] = [b]$, so $a \in [b]$, which means $a \sim b$ by definition of $[b]$.

3) Let $a, b \in A$. Want to show either $[a] = [b]$ or $[a] \cap [b] = \emptyset$.

There are two possibilities: either $[a] \cap [b] = \emptyset$ or $[a] \cap [b] \neq \emptyset$.

In the first case we are done.

So suppose $[a] \cap [b] \neq \emptyset$. Want to show $[a] = [b]$

Since $[a] \cap [b] \neq \emptyset$, there is some $c \in [a] \cap [b]$.

Since $c \in [a]$, we have $c \sim a$, so $a \sim c$ by reflexivity

Since $c \in [b]$, we have $c \sim b$.

Hence $a \sim b$ by transitivity

By (2), we get $[a] = [b]$.

□

Definition 2: Let A be non-empty set and $\mathcal{C}$ a collection of subsets of A.

We say that $\mathcal{C}$ is a partition of A if:

1) $\forall\ V \in \mathcal{C},\ V \neq \emptyset$

2) $\forall\ x \in A,\ \exists\ V \in \mathcal{C}$ such that $x \in V$. (ie $A = \bigcup_{V \in \mathcal{C}} V$)

3) $\forall\ V, W \in \mathcal{C}$, either $V = W$ or $V \cap W = \emptyset$. ("The elements are pairwise disjoint")

Examples: 1) $A = \mathbb{Z}$, $\mathcal{C} = \{E, O\}$ where $E = \{n \in \mathbb{Z} \mid n \text{ even}\}$, $O = \{n \in \mathbb{Z} \mid n \text{ odd}\}$

2) $A = \{a, b, c\}$, $\mathcal{C} = \{\{a\}, \{b, c\}\}$

3) $A = \mathbb{R}$, $\mathcal{C} = \{[x, x+1) \mid x \in \mathbb{Z}\}$



4) $A = [0,1] \subseteq \mathbb{R}$. Choose $N \in \mathbb{N}$. "Partition $[0,1]$ into N intervals of length $\frac{1}{N}$"

$\mathcal{C} = \{[0, \frac{1}{N}), [\frac{1}{N}, \frac{2}{N}), [\frac{2}{N}, \frac{3}{N}), \dots, [\frac{N-1}{N}, 1]\}$



Corollary 3 Let $\sim$ be an equivalence relation on a nonempty set A.

Let $\mathcal{C} = \{\text{Equivalence classes of } \sim\} = \{[a] \mid a \in A\}$.

Then $\mathcal{C}$ is a partition of A.

Corollary 4: Fix $n \in \mathbb{N}$. For $a \in \mathbb{Z}$, let $[a]$ denote its equivalence class modulo n.

Then 1) $\mathbb{Z} = [0] \cup [1] \cup \ldots \cup [n-1]$

2) For $j, k \in \{0, 1, \ldots, n-1\}$ if $j \neq k$ Then $[j] \cap [k] = \emptyset$.

Definition 5: Let $n \in \mathbb{N}$. We define the set of integers modulo n, denoted $\mathbb{Z}_n$, as the set of equivalence classes for the relation "congruence modulo n" on $\mathbb{Z}$.

So $\mathbb{Z}_n = \{[a] \mid a \in \mathbb{Z}\} = \{[0], [1], \ldots, [n-1]\}$.

Example: $\mathbb{Z}_5 = \{[0], [1], [2], [3], [4]\}$

$\mathbb{Z}_2 = \{[0], [1]\} = \{\text{Odds}, \text{Evens}\}$

Q: How do operations of $\mathbb{Z}$ replicate on $\mathbb{Z}_n$?

Well, for example, in $\mathbb{Z}_5$, we can try to define a sum $\oplus$ → \oplus

$[a] \oplus [b] = [a+b]$

eg. $[1] \oplus [2] = [1+2] = [3]$

but since $[1] = [6]$, we should have

$[1] \oplus [2] = [6] \oplus [2] = [8]$

So $[1] \oplus [2]$ can possibly give two different results.

However, that's not the case.

Theorem 6 Let $n \in \mathbb{N}$. Let $a, b, c, d \in \mathbb{Z}$. Then, in $\mathbb{Z}_n$, we have:

1) If $[a] = [b]$ and $[c] = [d]$, Then $[a+c] = [b+d]$

2) If $[a] = [b]$ and $[c] = [d]$, Then $[ac] = [bd]$

Proof: This is just a restatement of our Modular Arithmetic Thm (Thm 11.3)

Definition 7: Let $n \in \mathbb{N}$. By the previous Thm, on $\mathbb{Z}_n$ we can define operations of addition $\oplus$ (\oplus) and multiplication $\odot$ (\odot).

For $[a], [b] \in \mathbb{Z}_n$ we define

$$[a] \oplus [b] = [a+b] \qquad [a] \odot [b] = [a \cdot b]$$

Example: $\mathbb{Z}_5 = \{[0], [1], [2], [3], [4]\}$ ⇝ Math 402!!

$[1] \oplus [3] = [1+3] = [4]$, $[3] \oplus [3] = [6] = [1]$

$[4] \odot [2] = [8] = [3]$.